

ABIHASSAN K

Software Engineer | AI & Machine Learning | Python Developer

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PROFESSIONAL SUMMARY

Final-year Computer Science (AI & ML) graduate with hands-on experience in Python development, machine learning, computer vision, and automation. Delivered 3 internships and multiple end-to-end AI projects — including a real-time baby monitoring system and a multimodal AI vision assistant. Passionate about building scalable, intelligent software solutions. Seeking a Software Engineer role to contribute technical depth and a rapid learning mindset.

TECHNICAL SKILLS

Languages: Python, C

Frameworks & Libraries: TensorFlow, Keras, OpenCV, Mediapipe, Scikit-learn, Pandas, Matplotlib

Backend & Automation: Selenium, BeautifulSoup, Flask (basics), REST APIs

Databases: MySQL

Cloud & Tools: GitHub, Google Colab, Jupyter Notebook, VS Code, Arduino IDE

ML & AI: CNNs, RNNs, KNN, SVM, NLP, OCR, Computer Vision, Gesture Recognition

Software Practices: OOP, Agile basics, Data Structures & Algorithms, Version Control (Git)

WORK EXPERIENCE

Machine Learning Intern | HDLC Technology Pvt. Ltd. | *June 2024*

- Built real-time data dashboards using Pandas and Matplotlib, enabling live business insights.
- Applied classification algorithms (KNN, SVM) to real-world business datasets, improving data processing workflows.
- Developed and deployed data visualization pipelines for stakeholder reporting.

Machine Learning & AI Intern | Bharat Intern | *February 2024*

- Implemented real-time gesture tracking and face detection systems using OpenCV and Mediapipe.
- Integrated Google Speech Recognition API to build voice-activated command modules.

Web Scraping Developer Intern | MHR Connect Pvt. Ltd. | *February 2023*

- Built automated web scraping pipelines using Python, Selenium, and BeautifulSoup to extract structured data at scale.
- Stored and managed large unstructured datasets in MySQL with efficient schema design.
- Reduced manual data collection effort by automating multi-page scraping workflows.

PROJECTS

Smart Cradle with AI-Powered Baby Monitoring | | *March 2025*

- Engineered an IoT + AI solution with ESP32-CAM and OpenCV for real-time baby face tracking and monitoring.
- Implemented an audio classification model to detect infant crying sounds and trigger automated alerts.
- Integrated automated feeding controls, combining hardware (ESP32) and software (Python/OpenCV) seamlessly.

AI Vision Assistant — Gesture, OCR & Gender Detection | | *November 2024*

- Built a multimodal AI assistant using Python, Mediapipe, OpenCV, and deep learning (CNNs, Keras).
- Enabled real-time hand gesture recognition to trigger system commands with <100ms latency.
- Integrated live OCR to extract and read text from captured images via voice output.

- Implemented gender detection using facial recognition trained on a large dataset; achieved high classification accuracy.

Ornithopter Control System — Defence & Rescue | | *May 2024*

- Designed an agile aerial vehicle mimicking bird flight mechanics for surveillance and rescue operations.
- Implemented advanced flight control logic with real-time surveillance and victim detection capabilities.

EDUCATION

B.Tech Computer Science (AI & ML) — *CGPA: 8.2 | 2021–2025*

Hindustan Institute of Technology & Science, Chennai

CERTIFICATIONS

IBM Artificial Intelligence & Machine Learning Graduate — *July 2025 | Issued by IBM*

Verify: credly.com/badges/6eb151cb-1d1e-4a47-9f02-2bd45ca900fb

Deep Learning Specialization — *March 2025*

- CNNs, RNNs, backpropagation, optimization using TensorFlow and Keras; applied to image classification and NLP tasks.

Machine Learning Specialization — *December 2024*

- Supervised learning, decision trees, ensemble methods, bias/variance trade-offs using Python and Scikit-learn.